

Low-Cost LPG LEAKAGE DETECTOR

PAMARTHI KANAKARAJA

The circuit for an LPG leakage detector is readily available in the market, but it is extremely expensive and usually based on a microcontroller (MCU). Presented here is a low-cost circuit for an LPG detector that you can build easily.

The main objective of the circuit is to detect LPG leakage anywhere. Fig. 1 shows the author's prototype.

Circuit and Working

Circuit diagram of the low-cost LPG detector is shown in Fig. 2. It is built around step-down transformer X1, two rectifier diodes 1N4007 (D1 and D2), a 1000µF capacitor (C1), 7805 voltage regulator (IC1), MQ-6 LPG gas sensor (GS1), dual comparator LM393 (IC2), darlington transistor TIP122 (T2), 12V high-gain

Test Points

Test point	Details
TP0	GND
TP1	Around 16V DC
TP2	5V DC
TP3	High during gas leakage

PARTS LIST

Semiconductors:

- IC1 - 7805, 5V voltage regulator
- IC2 - LM393 dual comparator
- D1, D2 - 1N4007 rectifier diode
- T1 - TIP122 npn darlington transistor

Resistors (all 1/4-watt, ±5% carbon, unless stated otherwise):

- R1, R3 - 1-kilo-ohm
- R2 - 15-ohm, 0.5W
- VR1, VR2 - 10-kilo-ohm potmeter

Capacitor:

- C1 - 1000µF, 25V electrolytic

Miscellaneous:

- CON1 - 2-pin connector terminal
- X1 - 230V AC primary to 12V-0-12V, 750mA secondary transformer
- PZ1 - 12V high-gain siren/buzzer
- GS1 - MQ-6 LPG sensor

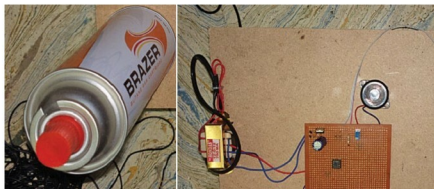


Fig. 1: Author's prototype

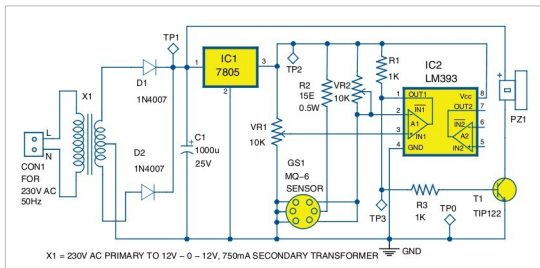


Fig. 2: Circuit diagram of the LPG detector

siren/buzzer (PZ1) and a few other components.

The mains supply is stepped down by transformer X1, rectified by a full-wave rectifier comprising diodes D1 and D2, filtered by capacitor C1 and fed to regulator 7805 (IC1) to maintain constant 5V DC output, which is fed to the circuit.

At the heart of